

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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SEC327 - R

Speed up cloud security investigations, response, and remediation

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Amazon Web Services



In this chalk talk we will cover

- 01 What is Amazon Detective
- 02 What is AWS Security Hub
- 03 How to integrate AWS Security Hub and Amazon Detective
- 04 How to speed up investigations, response, and remediation

Challenges security teams face



Lack of visibility
into security risks



Complexity of tools,
vendors, and processes



Too many alerts and not
enough security experts

What is Amazon Detective?



Amazon Detective simplifies the investigation process and helps security teams conduct **faster** and more **effective** investigations. With prebuilt summaries and context, you can quickly analyze and determine the extent of possible security issues.



Saving time and money by simplifying log collection



Automatic correlation of security findings to discover root cause



Threat hunting and searching for undetected threats

What is AWS Security Hub?



AWS Security Hub is a cloud security posture management service that **continuously** performs security best practice checks, **seamlessly** aggregates security findings from AWS and third-party services, and enables automated response.



Automated,
continuous best
practice checks



Consolidated
findings across
AWS services
and Partner
integrations



Standardized
findings format
and cross-Region
aggregation



Standards
aligned to
regulatory and
industry
compliance
frameworks



Automated
response,
remediation, and
enrichment
actions

Security Hub finding flows

Standards and controls



PCI DSS



NIST



AWS FSBP



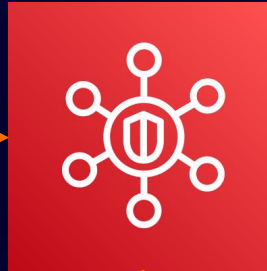
Center for Internet Security
Creating Confidence in the Connected World.

CIS AWS Foundations Benchmark



Service-managed standards

Security Hub



Amazon EventBridge

Remediation actions



AWS Lambda



AWS Systems Manager



AWS Step Functions

Findings ingestions



AWS Config



Amazon GuardDuty



Amazon Inspector



AWS Identity and Access Management (IAM)



Amazon Macie



AWS Firewall Manager



... and many more

Findings integrations



Amazon Detective



AWS Audit Manager



Amazon Security Lake



AWS Trusted Advisor



AWS Chatbot

Take action with AWS Partners



... plus many others



Speed up investigation, response, and remediation



Security Hub automation rules

Automation rules allow you to automatically update Security Hub findings based on criteria that you define, without any code.

Security Hub

Go to Aggregation Region

Summary

Controls

Security standards

Insights

Findings

Integrations

Automations New

Settings

What's new

Security Hub > Automations

Automations (5)

Edit Priority

Action

Create rule

Find rules

	Order	Name	Description	Status
☐	1	Suppress "[KMS.3] AWS KMS keys should not be deleted unintentionally" findings in dev and test accounts	Suppressing known informational issue from a control	Enabled
☐	2	Suppress findings for best practice control EC19 ingress for protected firewalled accounts	Suppressing false positive findings	Enabled
☐	3	Elevate finding severity for production accounts	Changing a high severity to critical for specific production accounts	Enabled
☐	4	Elevate severity for findings related to Crown Jewels resources	Setting finding's severity to Critical for specific finding's resource ID	Enabled
☐	5	Suppress informational findings from GuardDuty	Suppressing known informational issue from GuardDuty	Enabled

Security Hub automation rules

Use cases include:

- **Changing finding severity**
- **Suppressing findings**
- **Adding notes**

Security Hub > Automations > Create rule

Create rule

Rule Type

☒ Create a rule from template
use a pre-populated template for common scenario

☐ Create custom rule
Start with all blank fields

Rule template

Elevate severity for important resources

Elevate severity for important resources ✓

Suppress informational findings

Elevate severity for production accounts

Rule name

Elevate severity for important resources

The name must have 3 to 28 characters. Valid characters are a-z (lowercase only), 0-9, and - (hyphen)

Rule description

Elevate finding severity to Critical when specific S3 bucket at risk

Criteria

Choose which findings your rule applies to

Security Hub > Automations > Elevate severity for findings related to Crown Jewels resources

Elevate severity for findings related to Crown Jewels resources

Edit

Rule

Rule name	Rule description
Elevate severity for findings related to Crown Jewels resources	Setting finding's severity to Critical for specific finding's resource ID

Criteria

Key	Operator	Value
ResourceId	EQUALS	arn:aws:s3::examplebucket/developers/design_inf o.doc
Key	Operator	Value
WorkflowStatus	EQUALS	NEW

Automated action

Severity	Note
CRITICAL	Need urgent look into this critical S3 bucket

NEW

Amazon Detective automated investigations for AWS Identity and Access Management (IAM) entities

Amazon Detective now supports automated investigations of AWS Identity and Access Management (IAM) entities for indicators of compromise (IoC) to determine potential compromises and involvement in known tactics from the MITRE ATT&CK framework

By automatically investigating IAM entities for IOCs, Detective helps in identifying patterns and comprehending which resources are impacted by security events, offering a proactive approach to threat identification and mitigation.

Acting on findings



AWS Security Hub
Console



AWS Security Hub
Automation rules



Amazon Detective
Console



Amazon
EventBridge

+



AWS
Systems
Manager



AWS
Lambda



AWS Step
Functions

**DIY – build your
own automation**

Less technical expertise needed
Less flexibility

Higher technical expertise needed
Greater flexibility



AWS Security Hub
Custom actions

Demo



Security Hub automation rule

The screenshot displays the AWS Security Hub console interface. The top navigation bar includes the AWS logo, Services menu, Search bar, and user information. The left sidebar lists various navigation options, with 'Security Hub' selected. The main content area shows the 'Summary' page, which includes a filter bar and two widgets: 'Security standards' and 'Assets with the most findings'.

Security Hub Summary

Regions: All Linked Regions

Reset to default layout + Add widget

Choose a filter set Filter data

Workflow status = NEW Workflow status = NOTIFIED Record state = ACTIVE Clear filters

Security standards

Track your cloud security posture with a summary security score and standard security scores. This widget always shows complete, unfiltered data.

Security score

67%

Standard	Passed	Failed	Score
CIS AWS Foundations Benchmark v1.2.0	8	35	19%
AWS Foundational Security	175	62	74%

Assets with the most findings

Prioritize and evaluate your assets that are most at risk.

Resources Accounts Applications

Resources	By severity	By resource type
arn:aws:ec2:us-east-1:522536594833:instance/i-999999999		
arn:aws:eks:us-east-1:522536594833:cluster/GeneratedFindingEKSCl		

Detective finding groups

Resource Groups & Tag Editor

Detective

Summary

Finding groups

Investigations New

Search

Settings

Account management

Integrations New

General

Preferences

Usage

What's new

Getting started

Video tutorials

Detective > Finding groups > FindingGroup/0d1ebdd0-47e1-4a48-94df-c98fcd5a20da

TTPs (10), UnusualBehavior:AllAccounts (4) Info

High

Active

0d1ebdd0-47e1-4a48-94df-c98fcd5a20da

Related

Scope time Info

11/21/2023 16:00 - 11/21/2023 18:02 UTC

Summary - new Info

↻ Loading...

Observed tactics Info

Initial Access

Execution

Persistence

Privilege Escalation

Defense Evasion

Credential Access

Discovery

Lateral Movement

Collection

Command and Control

Exfiltration

Impact

Other

Visualization Info

This interactive visualization allows you to select and see details about each item in the finding group.

↻ Loading...

Involved entities

Involved findings

Entities Info

These entities are correlated to the same underlying activity. Pivot to an entity's profile to review additional details, or review the involved findings from the involved findings tab.

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Run Detective investigation

Resource Groups & Tag Editor

Detective

Summary

Finding groups

Investigations New

Search

Settings

Account management

Integrations New

General

Preferences

Usage

What's new

Getting started

Video tutorials

Detective > Search > AwsUser/AIDAXTKM3VWI4P6HPZIWC

ExternalUser

AWS user Info

Scope time Info

11/21/2023 08:00 UTC > 11/22/2023 08:00 UTC

Overview

New behavior

Resource interaction

Kubernetes activity

AWS user details Info

Run investigation

AWS user

ExternalUser

Created by

Admin

Finding groups

TTPs (10), UnusualBehavior:AllAccounts (4)

Principal ID

AIDAXTKM3VWI4P6HPZIWC

Created date

11/21/2023 17:43 UTC

AWS account

522536594833 (awsmorocco-sectech...

Last observed

11/21/2023 18:02 UTC

Investigations

Investigations generate an automatic report as an entry point for incident investigation. Data displayed is for all investigations, not based on scope time selection.

No investigations found

Try expanding your scope time, or run a new investigation

Findings associated with this resource Info

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Investigation report – Summary

The screenshot displays the AWS Detective console interface. On the left is a navigation sidebar with sections: **Detective** (containing Summary, Finding groups, Investigations *New*, and Search), **Settings** (containing Account management, Integrations *New*, General, Preferences, and Usage), and **What's new** (containing Getting started and Video tutorials). The main content area shows the investigation details for ID 170084144463811377517. At the top right of this area are **Archive** and **Download** buttons. The report title is **ExternalUser report summary** with an **info** icon and a red **High** severity badge. The summary text states: "We observed anomalous behavior for the user from 11/21/2023 - 11/22/2023 indicating potential compromise. The user invoked CloudTrail management actions mapped to Exfiltration and Impact MITRE tactic(s). The user was also involved in Findings that map to the MITRE tactic(s) Discovery, as well as other tactic(s). We've flagged 1 IP used by ExternalUser. The user was also involved in 4 findings, 1 finding group, 1 impossible travel, 2 new ASOs, 2 new geolocations, and 4 new user agents." Below this text is a table with three columns: **Scope time** (11/21/2023 08:00 UTC - 11/22/2023 08:00 UTC), **Indicators of compromise** (6 Tactics, 1 Flagged IP, 1 Impossible travel, 1 Finding group), and **Recommendation** (Based on our investigation, we recommend you take action to mitigate what we've found on AWS user ExternalUser. Please review [Security Best Practices in IAM](#) to secure your AWS resource.). Below the table are tabs for **Overview** and **Indicators**. The **Overview** tab is active, showing a section titled **IPs with high severity activity** with an **info** icon. A **Filter by IP** dropdown menu is set to 57.180.41.212. Below the filter is a diagram showing a central box for IP 57.180.41.212 with **Findings (4)**. This box is connected to three other boxes: **New geolocation Tokyo, JP**, **Impossible travel (1)**, and another box (partially obscured) representing a finding group.

Resource Groups & Tag Editor

Detective X

Detective > Investigations > 170084144463811377517

170084144463811377517

Archive Download

ExternalUser report summary info High

We observed anomalous behavior for the user from 11/21/2023 - 11/22/2023 indicating potential compromise. The user invoked CloudTrail management actions mapped to Exfiltration and Impact MITRE tactic(s). The user was also involved in Findings that map to the MITRE tactic(s) Discovery, as well as other tactic(s). We've flagged 1 IP used by ExternalUser. The user was also involved in 4 findings, 1 finding group, 1 impossible travel, 2 new ASOs, 2 new geolocations, and 4 new user agents.

Scope time	Indicators of compromise	Recommendation
11/21/2023 08:00 UTC - 11/22/2023 08:00 UTC	6 Tactics 1 Flagged IP 1 Impossible travel 1 Finding group	Based on our investigation, we recommend you take action to mitigate what we've found on AWS user ExternalUser. Please review Security Best Practices in IAM to secure your AWS resource.

Overview Indicators

IPs with high severity activity info

Filter by IP

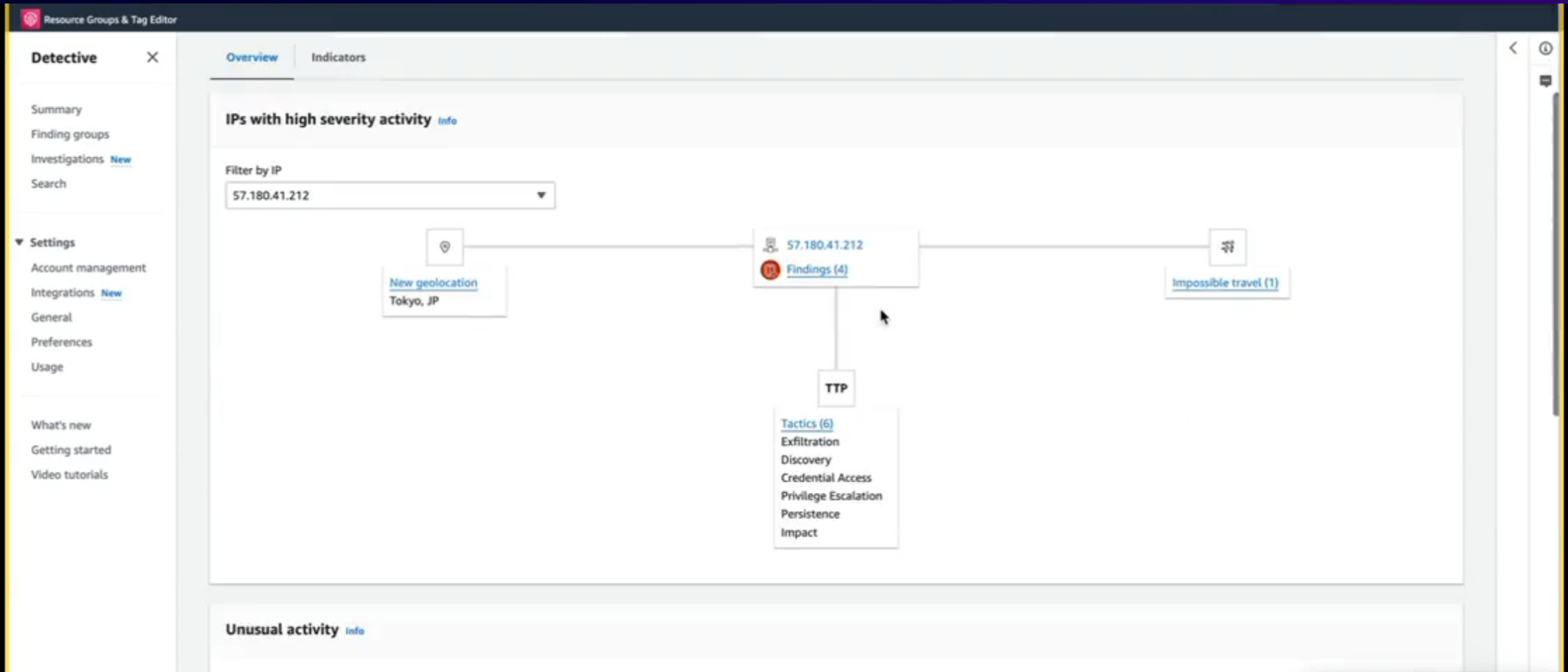
57.180.41.212

New geolocation Tokyo, JP

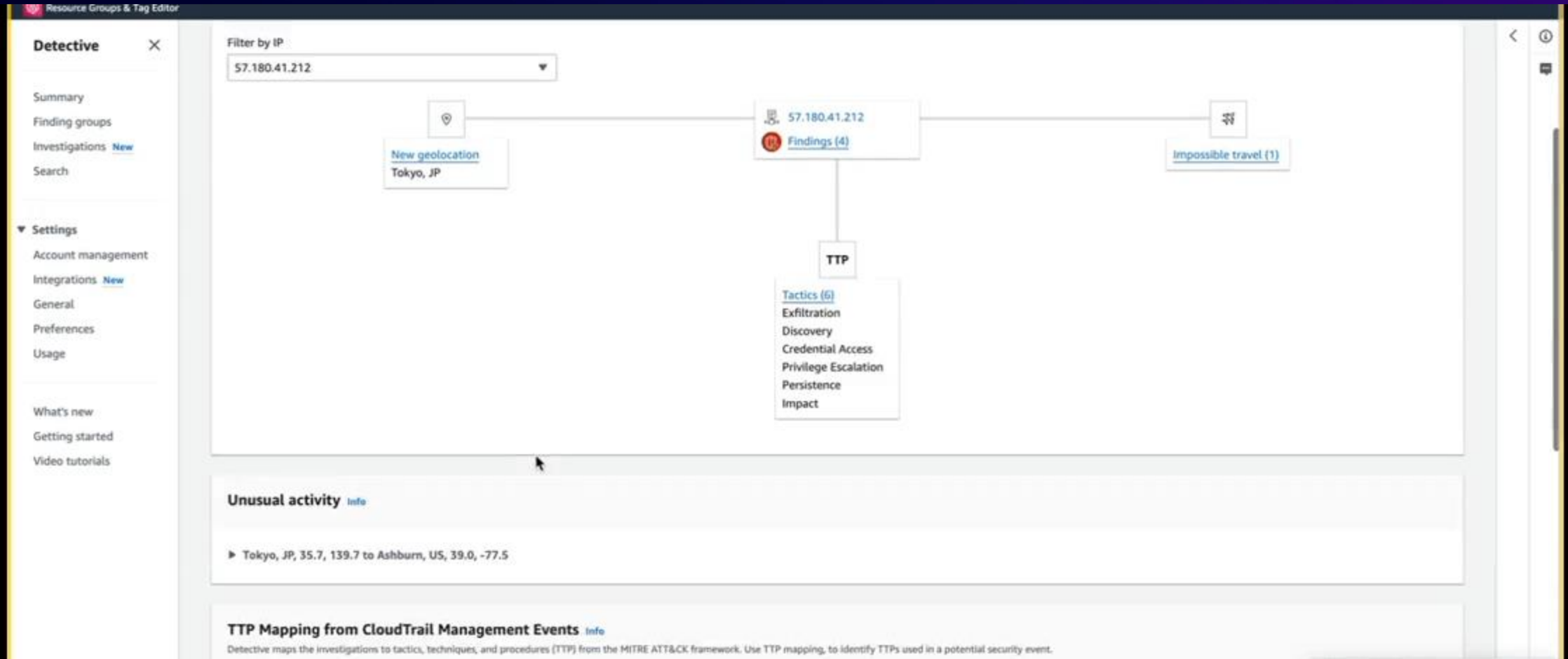
57.180.41.212 Findings (4)

Impossible travel (1)

Investigation report – Overview



Investigation report – TTPs



Raw logs Security Lake

Resource Groups & Tag Editor

Detective

Summary

Finding groups

Investigations **New**

Search

Settings

Account management

Integrations **New**

General

Preferences

Usage

What's new

Getting started

Video tutorials

Detective > Investigations > 170084628849121631242

170084628849121631242

Archive Download

ExternalUser report summary **Info** **High**

We observed anomalous behavior for the user from 11/21/2023 - 11/21/2023 indicating potential compromise. The user invoked CloudTrail management actions mapped to Exfiltration and Impact MITRE tactic(s). The user was also involved in Findings that map to the MITRE tactic(s) Discovery, as well as other tactic(s). We've flagged 1 IP used by ExternalUser. The user was also involved in 4 findings, 1 finding group, 1 impossible travel, 2 new ASOs, 2 new geolocations, and 4 new user agents.

Scope time

11/21/2023 16:00 UTC - 11/21/2023 19:00 UTC

Indicators of compromise

[6 Tactics](#)
[1 Flagged IP](#)
[1 Impossible travel](#)
[1 Finding group](#)

Recommendation

Based on our investigation, we recommend you take action to mitigate what we've found on AWS user ExternalUser. Please review [Security Best Practices in IAM](#) to secure your AWS resource.

Overview Indicators

IPs with high severity activity **Info**

Filter by IP

57.180.41.212

New geolocation
Tokyo, JP

57.180.41.212
Findings (4)

Impossible travel (1)

TTP

Tactics (6)
Exfiltration

Investigation report – Indicators

Resource Groups & Tag Editor

Detective

Summary

Finding groups

Investigations **New**

Search

Settings

Account management

Integrations **New**

General

Preferences

Usage

What's new

Getting started

Video tutorials

Detective > Investigations > 170084144463811377517

170084144463811377517

Archive Download

ExternalUser report summary Info High

We observed anomalous behavior for the user from 11/21/2023 - 11/22/2023 indicating potential compromise. The user invoked CloudTrail management actions mapped to Exfiltration and Impact MITRE tactic(s). The user was also involved in Findings that map to the MITRE tactic(s) Discovery, as well as other tactic(s). We've flagged 1 IP used by ExternalUser. The user was also involved in 4 findings, 1 finding group, 1 impossible travel, 2 new ASOs, 2 new geolocations, and 4 new user agents.

Scope time

11/21/2023 08:00 UTC - 11/22/2023 08:00 UTC

user

ExternalUser

Indicators of compromise

6 Tactics

1 Flagged IP

1 Impossible travel

1 Finding group

Recommendation

Based on our investigation, we recommend you take action to mitigate what we've found on AWS user ExternalUser. Please review [Security Best Practices in IAM](#) to secure your AWS resource.

Overview Indicators

Indicators (74) Info

Filter by Type, IP, Title and Severity

New geolocation

IP: 54.242.55.181 (Location: Ashburn, US, 39.0, -77.5)

New geolocation

IP: 57.180.41.212 (Location: Tokyo, JP, 35.7, 139.7)

Impossible travel

Origin Location: Tokyo, JP, 35.7, 139.7

Origin IP: 57.180.41.212

Destination Location: Ashburn, US, 39.0, -77.5

Destination IP: 54.242.55.181

Investigation report – Download

Resource Groups & Tag Editor

Detective

Summary

Finding groups

Investigations **New**

Search

Settings

Account management

Integrations **New**

General

Preferences

Usage

What's new

Getting started

Video tutorials

Detective > Investigations > 170084144463811377517

170084144463811377517

Archive Download

ExternalUser report summary Info High

We observed anomalous behavior for the user from 11/21/2023 - 11/22/2023 indicating potential compromise. The user invoked CloudTrail management actions mapped to Exfiltration and Impact MITRE tactic(s). The user was also involved in Findings that map to the MITRE tactic(s) Discovery, as well as other tactic(s). We've flagged 1 IP used by ExternalUser. The user was also involved in 4 findings, 1 finding group, 1 impossible travel, 2 new ASOs, 2 new geolocations, and 4 new user agents.

Scope time	Indicators of compromise	Recommendation
11/21/2023 08:00 UTC - 11/22/2023 08:00 UTC	6 Tactics	Based on our investigation, we recommend you take action to mitigate what we've found on AWS user ExternalUser. Please review Security Best Practices in IAM to secure your AWS resource.
user	1 Flagged IP	
ExternalUser	1 Impossible travel	
	1 Finding group	

Overview Indicators

Indicators (74) Info

Filter by Type, IP, Title and Severity

New geolocation

IP: 54.242.55.181 (Location: Ashburn, US, 39.0, -77.5)

New geolocation

IP: 57.180.41.212 (Location: Tokyo, JP, 35.7, 139.7)

Impossible travel

Origin Location: Tokyo, JP, 35.7, 139.7

Origin IP: 57.180.41.212

Destination Location: Ashburn, US, 39.0, -77.5

Destination IP: 54.242.55.181

aws

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Threat hunting

The screenshot displays the AWS Detective console interface. At the top, a green banner indicates a successful export of investigation 170084144463811377517. The left sidebar contains navigation links for Summary, Finding groups, Investigations (marked as 'New'), Search, Settings (Account management, Integrations, General, Preferences, Usage), What's new, Getting started, and Video tutorials. The main content area shows the investigation details for ID 170084144463811377517, titled 'ExternalUser report summary' with a 'High' severity tag. The summary text describes anomalous behavior from 11/21/2023 to 11/22/2023, mentioning CloudTrail actions, MITRE tactics, and 4 findings. Below the summary, three columns provide details: Scope time (11/21/2023 08:00 UTC - 11/22/2023 08:00 UTC), Indicators of compromise (6 Tactics, 1 Flagged IP, 1 Impossible travel, 1 Finding group), and Recommendation (Based on our investigation, we recommend you take action to mitigate what we've found on AWS user ExternalUser. Please review Security Best Practices in IAM to secure your AWS resource.). At the bottom, the 'Overview' tab is active, showing a section for 'IPs with high severity activity' with a filter by IP set to 57.180.41.212. A diagram below the filter shows a flow from a location icon to a box labeled '57.180.41.212 Findings (4)' and then to a findings icon.

Resource Groups & Tag Editor

Detective ×

Summary
Finding groups
Investigations **New**
Search

▼ Settings
Account management
Integrations **New**
General
Preferences
Usage

What's new
Getting started
Video tutorials

You have successfully exported 170084144463811377517. ×

Detective > Investigations > 170084144463811377517

170084144463811377517

Archive Download

ExternalUser report summary Info **High**

We observed anomalous behavior for the user from 11/21/2023 - 11/22/2023 indicating potential compromise. The user invoked CloudTrail management actions mapped to Exfiltration and Impact MITRE tactic(s). The user was also involved in Findings that map to the MITRE tactic(s) Discovery, as well as other tactic(s). We've flagged 1 IP used by ExternalUser. The user was also involved in 4 findings, 1 finding group, 1 impossible travel, 2 new ASOs, 2 new geolocations, and 4 new user agents.

Scope time	Indicators of compromise	Recommendation
11/21/2023 08:00 UTC - 11/22/2023 08:00 UTC	6 Tactics 1 Flagged IP 1 Impossible travel 1 Finding group	Based on our investigation, we recommend you take action to mitigate what we've found on AWS user ExternalUser. Please review Security Best Practices in IAM to secure your AWS resource.

Overview Indicators

IPs with high severity activity Info

Filter by IP
57.180.41.212 ▼

57.180.41.212 Findings (4)

```

    "IndicatorDetail": {
      "RelatedFindingDetail": {
        "Arn": "arn:aws:guardduty:us-east-1:522536594833:detector/80b183dfd1f8e934120c4e04df20ce26/finding/d0c5f98593dcad837cf0ff54f3f44f
04",
        "Type": "TTPs/UnauthorizedAccess:IAMUser-MaliciousIPCaller.Custom",
        "IpAddress": "57.180.41.212"
      }
    }
  },
  {
    "IndicatorType": "RELATED_FINDING",
    "IndicatorDetail": {
      "RelatedFindingDetail": {
        "Arn": "arn:aws:guardduty:us-east-1:522536594833:detector/80b183dfd1f8e934120c4e04df20ce26/finding/fac5f985932bd6784172f47c45ce35
4d",
        "Type": "TTPs/Discovery/Recon:IAMUser-MaliciousIPCaller.Custom",
        "IpAddress": "57.180.41.212"
      }
    }
  },
  {
    "IndicatorType": "RELATED_FINDING",
    "IndicatorDetail": {
      "RelatedFindingDetail": {
        "Arn": "arn:aws:guardduty:us-east-1:522536594833:detector/80b183dfd1f8e934120c4e04df20ce26/finding/86c5f9830ee48d331d96a930278ab9
c2",
        "Type": "TTPs/Discovery:S3-MaliciousIPCaller.Custom",
        "IpAddress": "57.180.41.212"
      }
    }
  },
  {
    "IndicatorType": "FLAGGED_IP_ADDRESS",
    "IndicatorDetail": {
      "FlaggedIpAddressDetail": {
        "IpAddress": "57.180.41.212",
        "Reason": "AWS_THREAT_INTELLIGENCE"
      }
    }
  }
]
}

```

```

(python-sdk-test) awsmorocco-sectech-Admin c889f3bd5359:8 tanashr$ aws detective list-indicators --graph-arn 'arn:aws:detective:us-east-1:5225365948
33:graph:fdac8011456e4e6182facb26dfceade0' --investigation-id '170084941141317305011'

```

Thank you!



Please complete the session
survey in the mobile app

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